MIND MIRROR

CHECK IN. REFLECT. GROW.

# AI JOURNAL WITH SENTIMENT ANALYSIS

**SPRINT 2**

17 April, 2026

# AGENDA OVERVIEW

# AGENDA OVERVIEW CONTD.

# TEAM MEMBERS

**KRISHNA**
Scrum Master | Developer

**CHINTAN KOTADIYA**
Developer | Tester

**RAFAELA HOLLANDA**
Team Leader | Developer

**VANESSA AGBUGBA**
Front End Developer

**SANKET MATROJA**
AI Generalist

**SAI**
Developer

# IMPROVEMENTS MADE FROM PROFESSOR'S FEEDBACK

**Feedback:** Link on wikipage was not accurate
**Action:** Wikipage was updated with accurate links

# PROJECT DESCRIPTION

| | |
|---|---|
| **Project Name:** | Mind Mirror |
| **Team:** | Nexus |
| **Project Description:** | For individuals who struggle to understand and manage their emotions, the AI-powered journaling application is a smart, emotion-tracking tool that provides real-time sentiment analysis, mood visualization, and actionable insights. Unlike traditional journaling apps that lack emotional analytics or structure, our application combines journaling with AI-driven insights to help users recognize patterns, prevent burnout, and improve emotional well-being. |
| | |
| **Benefit Outcomes:** | Users can anticipate measurable improvements in emotional awareness, reflection consistency, and stress management. The application is expected to increase emotional recognition by 60%, reduce stress levels by 40%, and triple journaling consistency through its quick-entry and real-time feedback features. |
| **Github Link:** | https://github.com/htmw/2026S-Nexus |

# TEAM WORKING AGREEMENT

## COMMUNICATION PLAN

- Primary instant meeting platform: WhatsApp
- Video Meeting: Discord

Team members are expected to:

- Communicate respectfully and professionally
- Respond to messages within a reasonable time frame (ideally within 24 hours)
- Share progress updates, blockers, and concerns openly
- Use the agreed communication tools for all project-related discussions

## MEETINGS & AVAILABILITY

The team will hold three regular sync meetings per week:

- Sunday: 10:00 AM • Tuesday: 6:30 PM • Thursday: 9:00 PM

Additional meetings may be scheduled as needed based on sprint requirements. Expectations:

- All members should attend scheduled meetings
- If unable to attend, members must notify the team in advance
- Meeting notes and action items will be documented and shared

# TEAM WORKING AGREEMENT

## TASK MANAGEMENT

- Task tracking tool: Jira

All project tasks will be:

- Clearly defined and assigned
- Tracked through Jira
- Updated regularly to reflect progress
- Completed before agreed internal deadlines

## CONFLICT RESOLUTION

In the event of conflict:

1. The concerned party should first express the issue to the team openly and respectfully
2. The team will attempt to resolve the issue collaboratively
3. If unresolved, the issue will be escalated to the Team Leader / Scrum Coordinator
4. Further escalation (e.g., to the instructor) will be considered only if necessary

# TEAM WORKING AGREEMENT

## WORK EXPECTATIONS & DEFINITION OF DONE

Each task is considered “Done” when:

- It meets the agreed acceptance criteria
- It is reviewed by at least one other team member (when applicable)
- Code runs without critical errors
- Relevant documentation is updated
- Changes are committed to the shared repository

Team members are responsible for:

- Completing assigned work on time
- Communicating blockers early
- Supporting teammates when needed

# STUDENT PERSONA

## NAME: SARAH MARTINEZ

**Age**: 21

**Gender:** Female

**Occupation:** Undergraduate Computer Science Student

Sarah is a full-time college student balancing coursework, part-time work, and social commitments. During exam periods, her stress levels rise significantly, but she often ignores the early warning signs of burnout. She occasionally journals but finds long entries overwhelming and inconsistent. She wants a simple way to reflect daily without spending too much time.

Challenges:

- Experiences fluctuating stress levels during the semester
- Inconsistent journaling habits
- Difficulty recognizing emotional patterns over time
- Feels overwhelmed during midterms and finals

Goals:

- Track emotional trends throughout the semester
- Identify early signs of burnout
- Receive short, actionable suggestions
- Maintain better emotional balance during academic pressure

# WORKING PROFESSIONAL PERSONA

## NAME: DAVID CHEN

**Age:** 32

Gender: Male

**Occupation:** Software Engineer

David works remotely for a tech company and spends most of his day attending meetings and coding. While he performs well professionally, he often feels mentally drained without clearly understanding the cause. He values productivity and self-improvement but rarely pauses to reflect on his emotional state.

Challenges:

- Blurred boundaries between work and personal life
- Emotional fatigue during high-pressure deadlines
- No structured way to monitor mood patterns
- Limited time for long reflection exercises

Goals:

- Understand how work stress affects his mood
- Monitor emotional stability over time
- Receive quick, data-driven insights
- Improve work-life balance

# WORKING PARENT PERSONA

## NAME: EMILY PETERSON

**Age:** 40

**Gender:** Female

**Occupation:** HR Manager

Emiliy balances a demanding job with parenting responsibilities. Her schedule leaves little time for self-reflection, yet she recognizes the importance of emotional awareness. She often feels emotionally exhausted but cannot clearly identify triggers or patterns. She prefers simple, easy-to-use digital tools.

Challenges:

- Limited time for lengthy journaling
- Emotional fatigue from multitasking responsibilities
- Difficulty identifying emotional triggers
- Wants non-clinical, supportive guidance

Goals:

- Build daily emotional awareness
- Identify patterns in stress levels
- Receive manageable well-being suggestions
- Maintain emotional stability for herself and her family

# SELF-IMPROVEMENT ENTHUSIAST PERSONA

## NAME: ALEX RIVERA

**Age:** 27

**Gender:** Male

**Occupation:** Fitness Coach

Alex actively tracks his workouts, sleep cycles, and productivity habits. He believes emotional well-being is just as important as physical health but lacks a structured system to measure it. He is interested in data-driven insights and enjoys tools that provide analytics and visualizations.

Challenges:

- Journaling apps provide no analytical insights
- No measurable way to track emotional consistency
- Wants deeper pattern recognition
- Prefers technology-backed recommendations

Goals:

- Quantify emotional trends
- Compare mood ratings with AI sentiment analysis
- Identify emotional volatility
- Optimize overall mental performance

# TECHNOLOGIES

## FRONT END

React

JS

HTML

CSS

## BACK END

FastAPI

uvicorn

PyTorch

Hugging Face

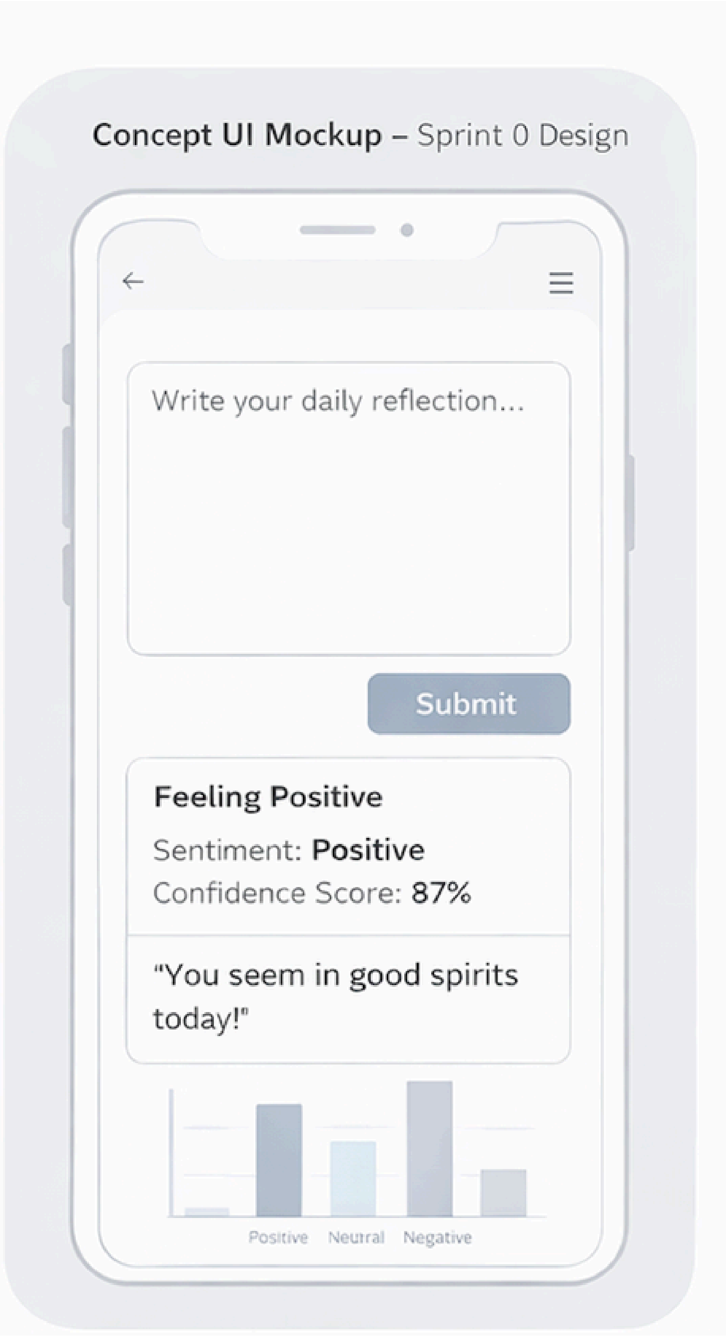
## DATABASE

mongoDB

## TOOLS

Canva

# MINIMAL VIABLE PRODUCT (MVP)



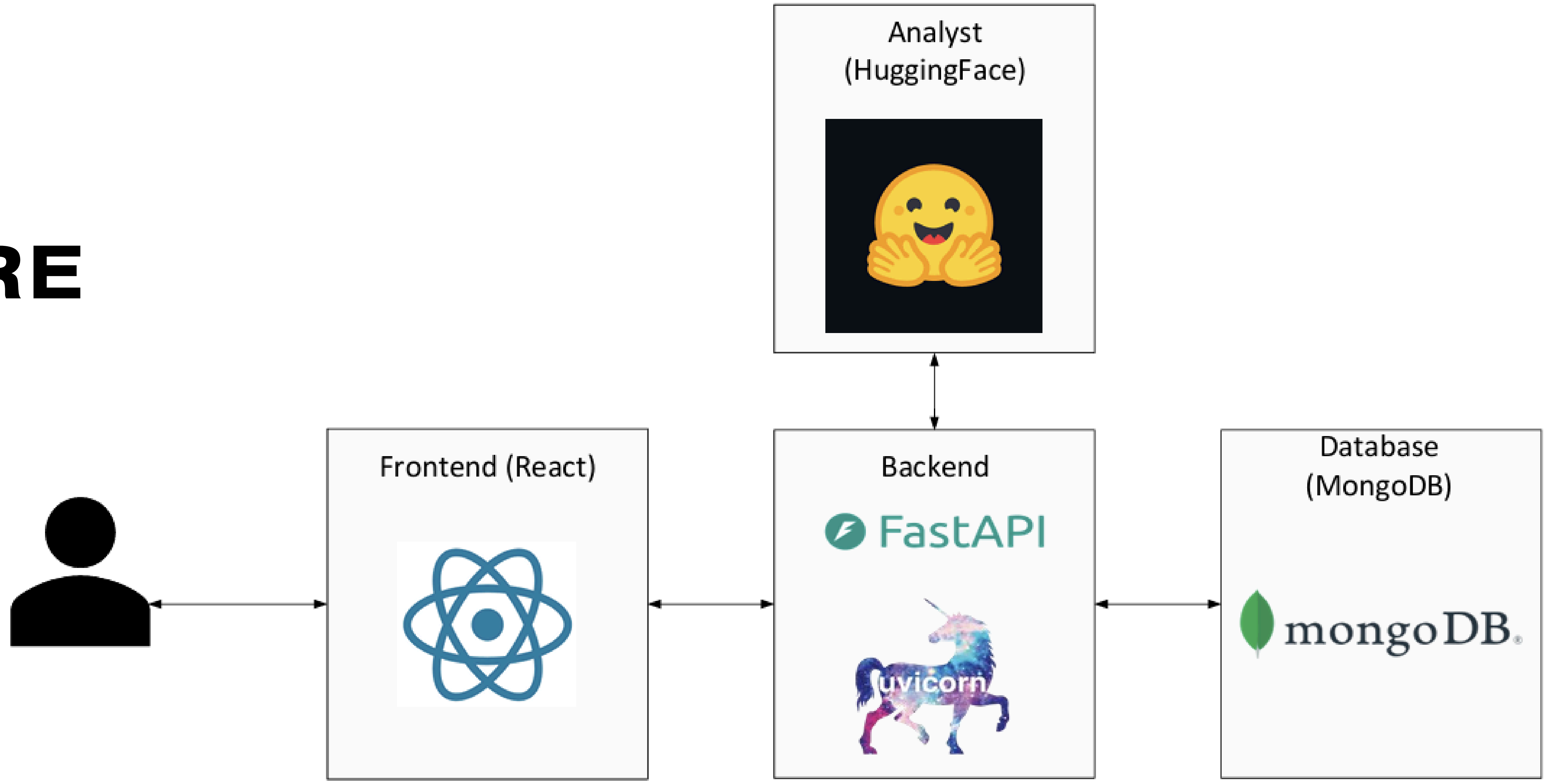

An AI-powered journaling application with real-time emotional analysis and visual mood tracking.

- **Real-Time Sentiment Analysis** – Journal entries are analyzed instantly using an NLP model.
- **Mood Classification System** – Each entry is labeled as Positive, Neutral, or Negative.
- **Confidence Score Display** – Shows the model's certainty level for each classification.
- **Immediate Mood Feedback** – Provides short, supportive emotional insights after analysis
- **Visual Mood Distribution** – Displays a basic chart showing mood trends over time.
- **Local Entry Persistence** – Saves journal entries locally for later review (MVP scope).

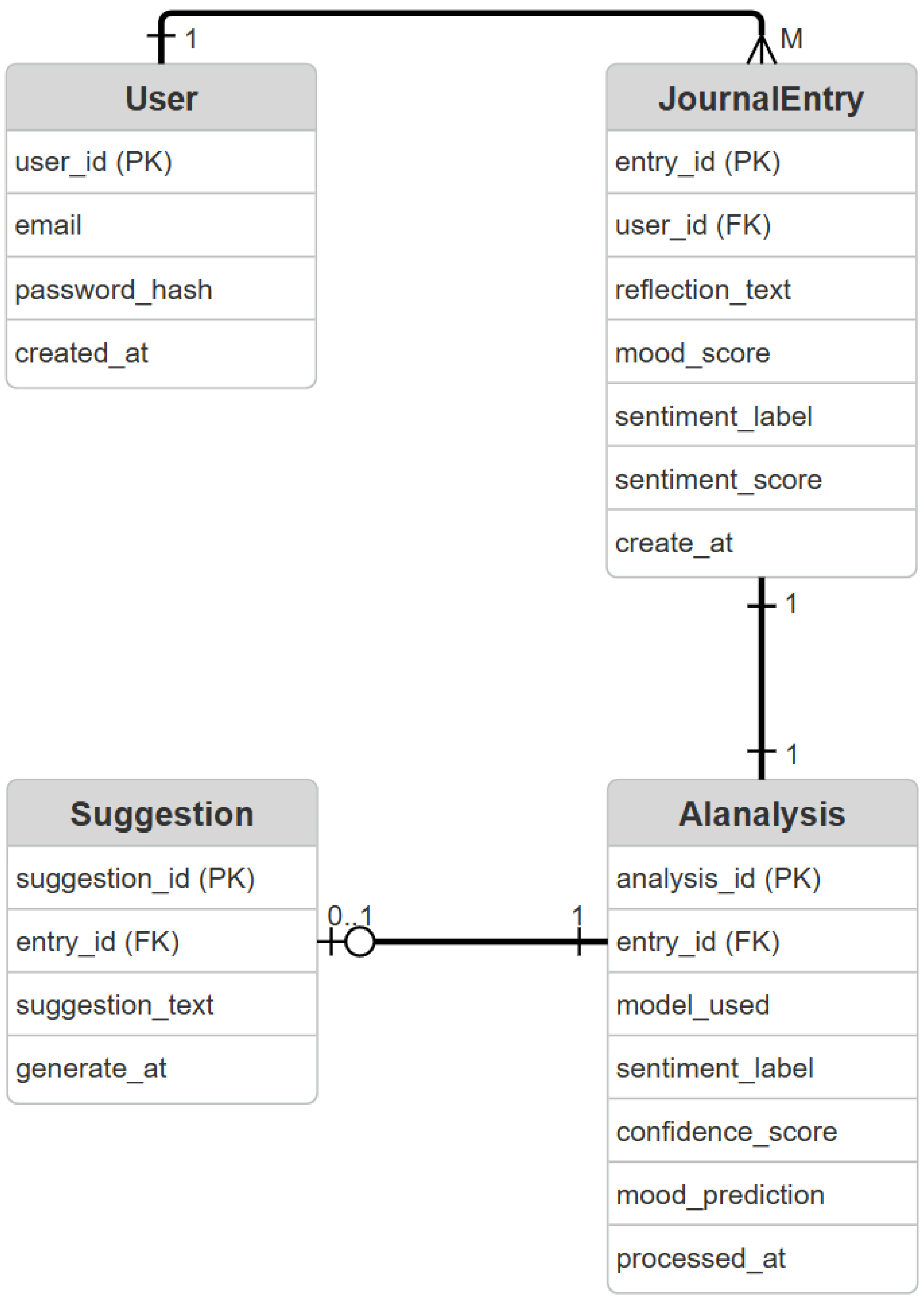
# COMPONENTS

- **Journal Input Manager :** Handles user journal entry input and submission through the web interface.
- **Backend API Manager :** Manages HTTP requests and communication between frontend, sentiment model, and database.
- **Sentiment Analysis Manager :** Processes journal text and classifies sentiment (Positive / Neutral / Negative).
- **Confidence Scoring Module:** Extracts prediction probability scores from the DistilBERT output.
- **Mood Feedback Generator:** Converts sentiment results into short, supportive emotional insights.
- **Data Persistence Layer :** Stores journal entries, sentiment labels, confidence scores, and timestamps.
- **Mood Visualization Manager:** Retrieves stored data and generates mood distribution charts over time.

# ARCHITECTURE DIAGRAM

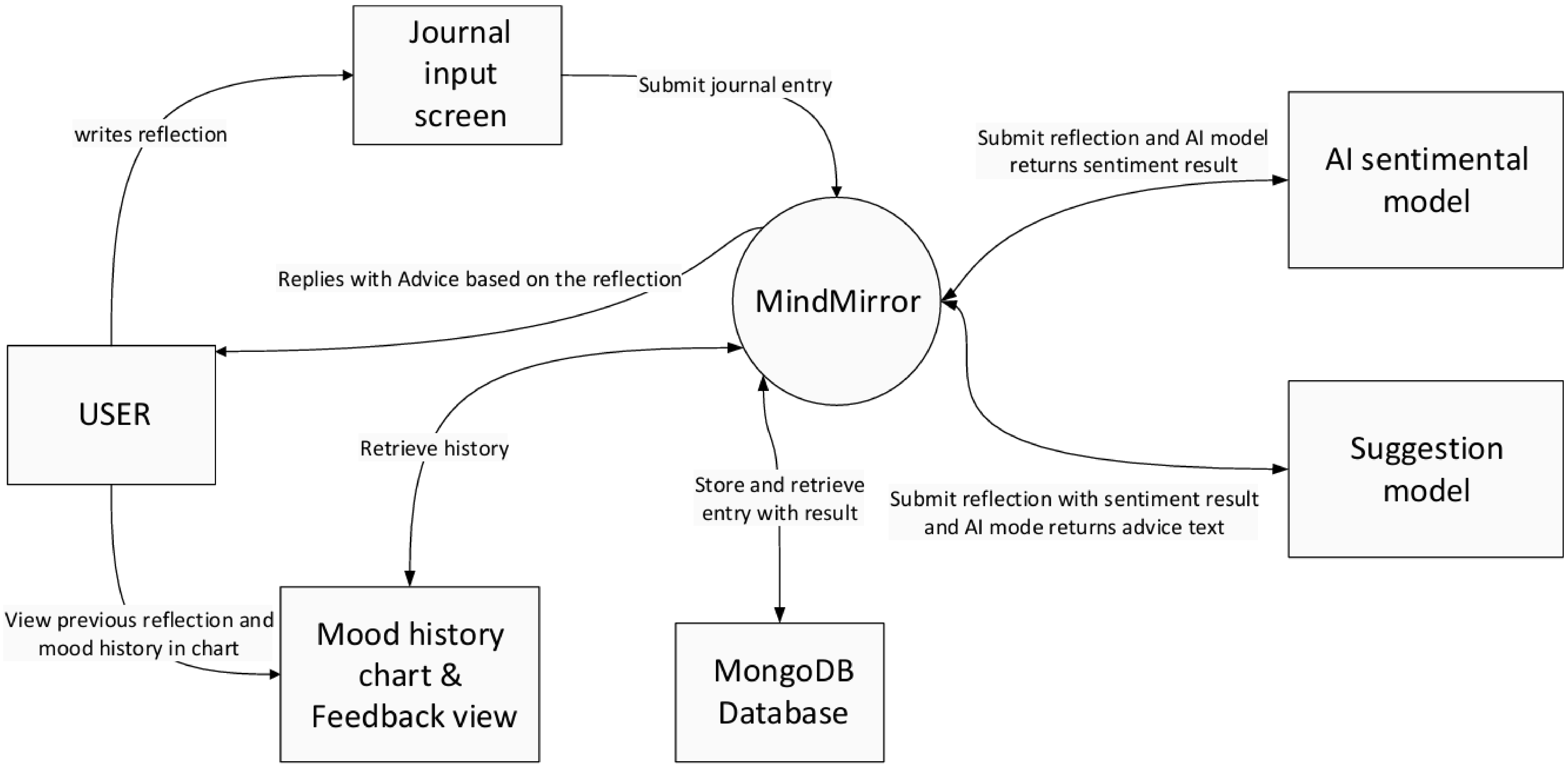


# ER DIAGRAM

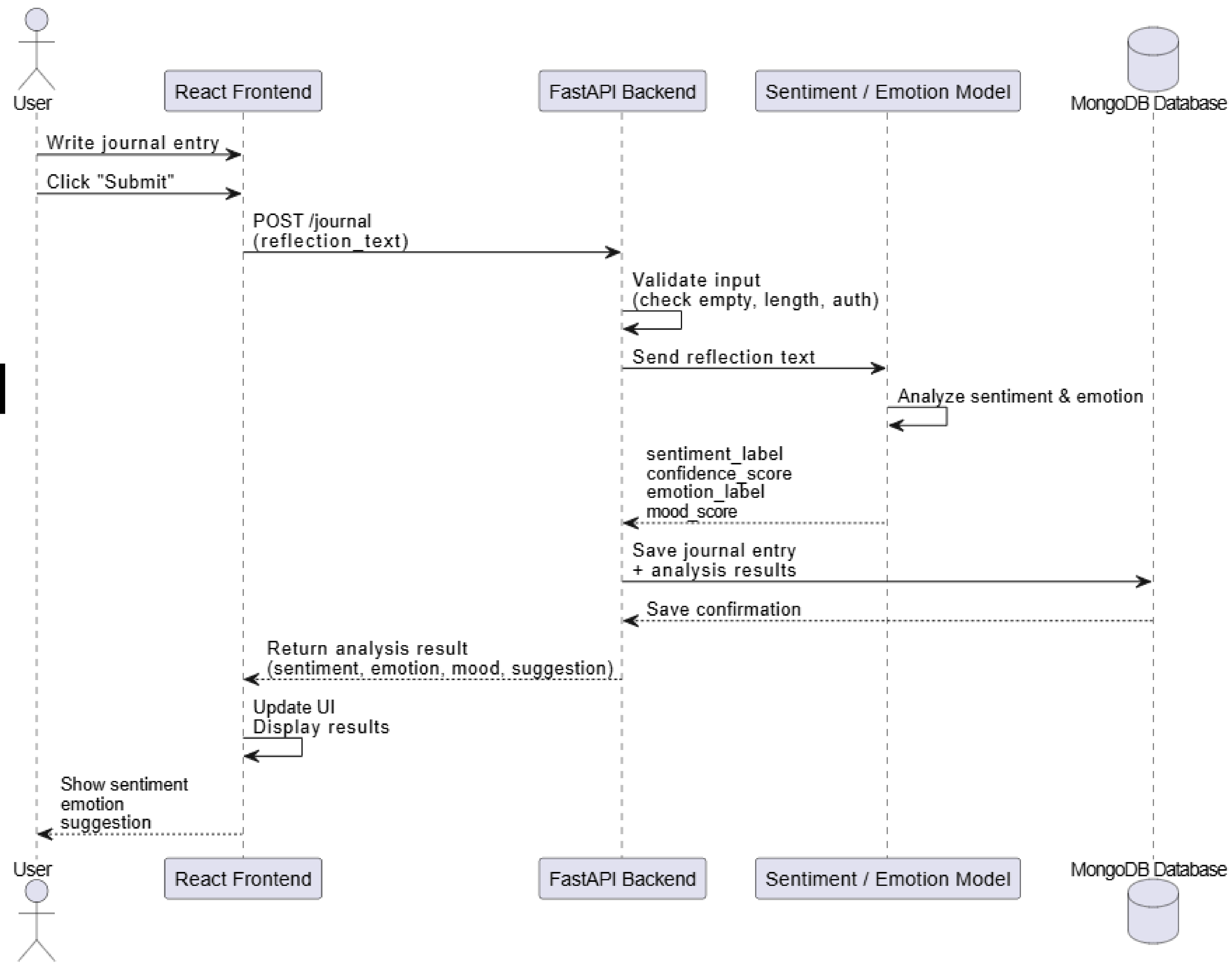


1
M
User
user_id (PK)
email
password_hash
created_at
JournalEntry
entry_id (PK)
user_id (FK)
reflection_text
mood_score
sentiment_label
sentiment_score
create_at
1
1
Suggestion
suggestion_id (PK)
entry_id (FK)
suggestion_text
generate_at
0..1
1
AIanalysis
analysis_id (PK)
entry_id (FK)
model_used
sentiment_label
confidence_score
mood_prediction
processed_at

# CONTEXT DIAGRAM

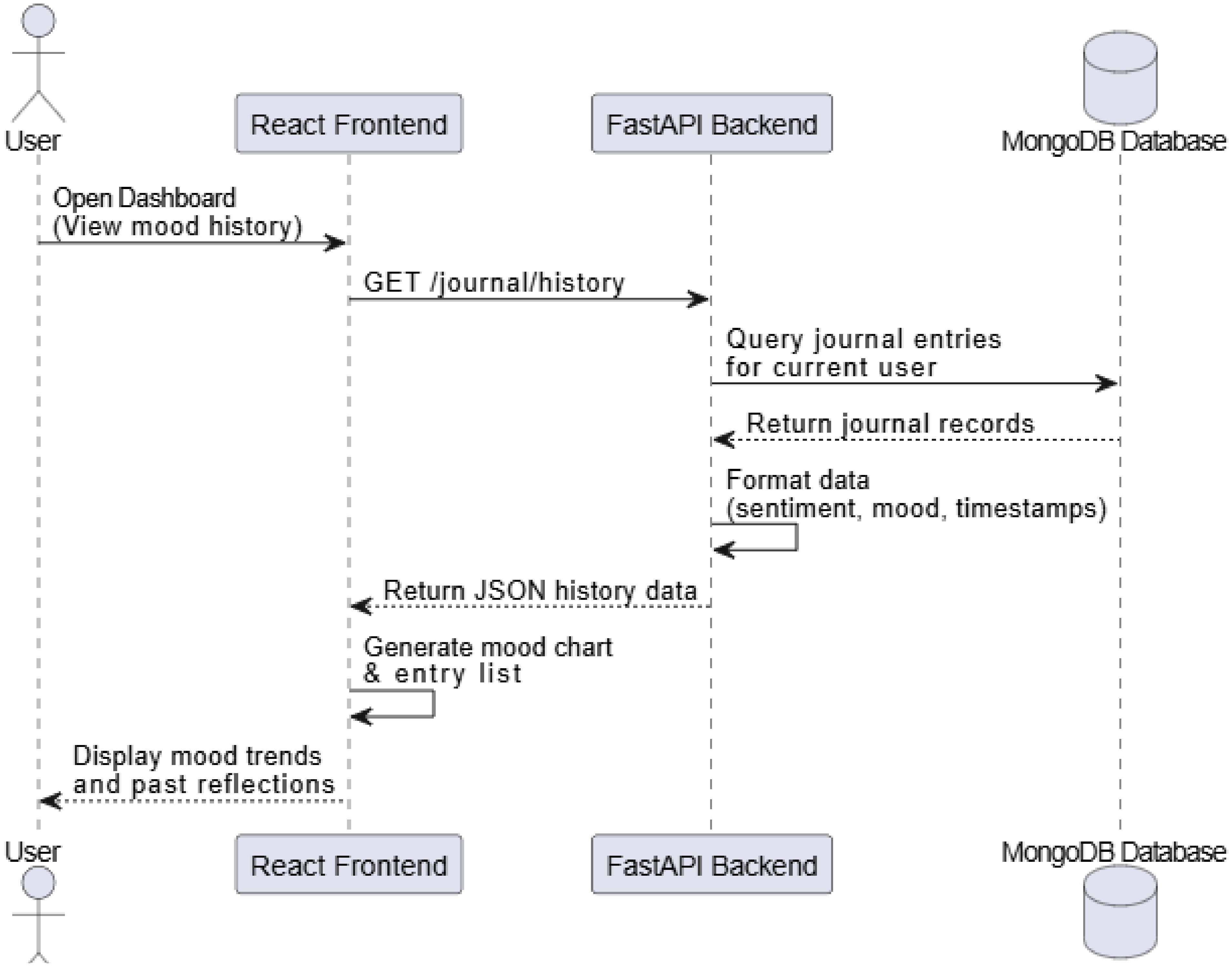


# SEQUENCE DIAGRAM -JOURNAL ENTRY




Team Nexus

# SEQUENCE DIAGRAM -DASHBOARD



# SPRINT 1 RECAP

## What We Set Out To Do

We aimed to build a complete journaling experience where users can submit entries and have them processed through AI sentiment analysis. This included displaying sentiment results along with confidence scores, while ensuring a smooth user experience through proper input validation, submission confirmation, error handling, and automatic input reset. Additionally, we planned to allow users to view their previous journal entries (US13).

## What We Delivered

- **US1, US7, US10: Core Journal Functionality:** Essential features that allow users to create, access, and manage their journal entries.
- **US2, US3, US4:** Processing and AI Analysis: Journal Entry Processing, AI Sentiment Analysis and confidence score.
- **US5, US6, US8:** Results and User Feedback: Sentiment Result Display, mood visualization and submission confirmation.

# PRODUCT BACKLOG

## CORE JOURNAL FUNCTIONALITY

**US11 – User Registration**
As a user, I want to register an account so that my journal entries are securely stored.
Acceptance Criteria
- Email & password required
- Password hashed
- User saved in database
- Duplicate email prevented

**Priority:** Must **Story Points**: 5

**US12 – User Login**
As a user, I want to log in so that I can access my journal dashboard.
Acceptance Criteria
- JWT token generated
- Protected routes enforced
- Login error handling

**Priority:** Must **Story Points**: 5

**US13 – View Journal Entries**
As a user, I want to view my previous journal entries so that I can reflect on past emotional states.
Acceptance Criteria
- List view of entries
- Emotion label shown
- Sorted by date
- Entries linked to user ID
- Users access only their own data

**Priority:** Must **Story Points**: 8

## RESULTS & USER FEEDBACK

**US6 – Mood Visualization**
As a user, I want to see a simple visualization of my mood history so that I can begin identifying patterns.
Acceptance Criteria
- Simple chart renders
- Updates after submission

**Priority:** Should **Story Points**: 5

**US16 – Error Handling**
As a user, I want reliable error handling so that I understand when something goes wrong.
Acceptance Criteria
- API errors handled gracefully
- Clear error messages displayed
- No system crashes

**Priority:** Must **Story Points**: 3

**US18 – Improved Mood Visualization**
As a user, I want a clearer visualization of my mood data so that I can better understand trends.
Acceptance Criteria
- Chart labels are clear
- Axes properly defined
- Colors distinguish sentiment categories

**Priority:** Should **Story Points**: 3

## PROCESSING & AI ANALYSIS

**US15 – Store Sentiment Results**
As a system, I want to store sentiment and confidence results so that they can be used for future analysis and visualization.
Acceptance Criteria
- Sentiment label stored in database
- Confidence score stored
- Data retrievable for visualization and history

**Priority:** Should **Story Points**: 5

Detailed spreadsheet here

# PRODUCT BACKLOG

## ADDITIONAL FEATURES

**US19 – Secure Backend Endpoints**
As a system, I want to secure backend endpoints so that only authenticated users can access protected resources.
Acceptance Criteria
- JWT required
- Unauthorized requests blocked
- Proper error responses

**Priority:** Must **Story Points**: 3

## FUTURE FEATURES

**US20 – Therapist Access**
As a user, I want my therapist to have access to my journal and mood data.

**Priority:** Could **Story Points**: 5

**US21 – Enable Sharing**
As a user, I want to enable sharing of my journal data.

**Priority:** Could **Story Points**: 8

**US22 – Disable Sharing**
As a user, I want to disable sharing at any time.

**Priority:** Could **Story Points**: 5

**US23 – Therapist User List**
As a therapist, I want to see my assigned users

**Priority:** Could **Story Points**: 5

**US24 – Therapist Dashboard**
As a therapist, I want to view journal entries and mood trends.

**Priority:** Could **Story Points**: 5

Detailed spreadsheet here

# SPRINT 2 BACKLOG

**Sprint Name:** Sprint 2

**Sprint Duration:** April 3rd - April 16th

**Sprint Length:** 2 Weeks

**Sprint Goal**
Incorporate users into the system by implementing secure registration, login, and authenticated access to journal features.

**Stories Committed**
US6, US11, US12, US13, US15, US16, US17, US18, US19.

**Total Story Points Committed: 40**



Detailed spreadsheet here

# SPRINT 2 TASK BREAKDOWN

During Sprint 2, user stories were decomposed into technical tasks to distribute work among team members and track implementation progress.

Task breakdown allows the team to:

- Assign clear responsibilities
- Estimate development effort
- Track sprint progress
- Enable parallel development

| US1 1- Design Registration Form | | | | |
|---|---|---|---|---|
| **Item** | **Status** | **Owner** | **Due Date** | **Est. Hours** |
| Build a registration page with fields | Com... | VA Vanessa Ag... | 15 Apr ... | 4 |
| Write tests for registration flow | Com... | Chintan Ko... | 15 Apr ... | 1 |
| Test every protected endpoint | Com... | | 15 Apr ... | 2 |

| US15 - Add Sentiment Fields to JournalEntry schen | | | | |
|---|---|---|---|---|
| **Item** | **Status** | **Owner** | **Due Date** | **Est. Hours** |
| Add columns to JournalEntry | Com... | | 15 Apr ... | 3 |
| Verify that after sentiment analysis, both label and score are saved to DB | Com... | | 15 Apr ... | 2 |
| Create an Express/FastAPI global error handling | Com... | | 15 Apr ... | 2 |



Detailed spreadsheet here

# SPRINT 2 TEST CASE

**Sprint 2**

**US16 — As a user, I want reliable error handling so that I understand when something goes wrong during submission or analysis.**

**Acceptance criteria:** *Backend/model errors caught; clear error displayed; no crash on ML failure; safe 500 responses returned*

| TC ID | Scenario | Type | Status | Description | Input / Payload | Expected Result | Notes |
|---|---|---|---|---|---|---|---|
| TC_US16_01 | System response | Integration | PASS | Network timeout occurs during submission and frontend maps it to a friendly message | Simulated timeout / backend unreachable | Friendly error shown: Could not reach the server. Please try again. | Frontend interceptor check |
| TC_US16_02 | System response | Integration | PASS | ML or sentiment analysis fails during journal submission | Valid entry + ML service throws error | HTTP 200; entry still saved; warning message returned | No-crash requirement |
| TC_US16_03 | Invalid input | Integration | PASS | User submits malformed or invalid payload | `{"mood":99} or missing required field` | HTTP 400; clear JSON error and code returned | Validation handling |
| TC_US16_04 | System response | Unit / Integration | PASS | Database failure occurs during save | Valid payload + mocked/broken DB connection | HTTP 500; safe message returned; no internal details leaked | Safe server failure |
| TC_US16_05 | System response | Integration | PASS | Unknown route or unexpected server error is triggered | Invalid route / forced exception | HTTP 404 or 500; consistent JSON error shape returned | Global error handler check |
| **Totals** | | | 5 | 0 | 0 | | |

Detailed spreadsheet here

# SPRINT 2 TEST CASE

**Sprint 2**

**US12 — As a user, I want to log in securely so that I can access protected features.**

**Acceptance criteria:** *Valid login returns token/session; wrong credentials rejected; protected routes require valid JWT; missing token rejected*

| TC ID | Scenario | Type | Status | Description | Input / Payload | Expected Result | Notes |
|---|---|---|---|---|---|---|---|
| TC_US12_01 | Valid flow | Integration | PASS | User logs in with valid email and password | `{"email":"sanket.test@example.com","password":"Password123!"}` | HTTP 200; JWT/token returned; dashboard access allowed | Happy path login |
| TC_US12_02 | Invalid input | Integration | PASS | User logs in with wrong password | Known email + wrong password | HTTP 401; generic login failure message | No credential leakage |
| TC_US12_03 | Invalid input | Integration | PASS | Unknown email attempts login | Unknown email + any password | HTTP 401; generic login failure message | User enumeration avoided |
| TC_US12_04 | System response | Integration | PASS | Protected route called without token | No Authorization header | HTTP 401; request rejected | JWT middleware check |
| TC_US12_05 | System response | Integration | PASS | Protected route called with invalid or expired token | Bad token / expired token | HTTP 401; request rejected | Feature verified; dedicated auth test task listed In Progress |
| **Totals** | | | 5 | 0 | 0 | | |

Detailed spreadsheet here

# SPRINT 2 TEST CASE

**Sprint 2**

**US11 — As a user, I want to register for an account so that I can securely use the application.**

**Acceptance criteria:** *Registration form validates inputs; duplicate email blocked; password stored hashed; account created successfully*

| TC ID | Scenario | Type | Status | Description | Input / Payload | Expected Result | Notes |
|---|---|---|---|---|---|---|---|
| TC_US11_01 | Valid flow | Integration | PASS | User registers with valid email and password | `{"email":"sanket.test@example.com","password":"Password123!"}` | HTTP 201; user created successfully | Happy path registration |
| TC_US11_02 | Invalid input | Integration | PASS | User tries to register with duplicate email | Existing email reused | HTTP 409; clear duplicate email message returned | Duplicate account prevention |
| TC_US11_03 | Invalid input | Integration | PASS | User submits invalid email or missing fields | Malformed or incomplete payload | HTTP 400/422; field validation error shown | Server-side validation |
| TC_US11_04 | System response | Integration | PASS | Verify password is not stored in plaintext | Successful registration record in DB | Password saved as hash not raw password | Feature verified; dedicated auth test task listed In Progress |
| **Totals** | | | 4 | 0 | 0 | | |

Detailed spreadsheet here

# SPRINT 2 TEST CASE

**Sprint 2**

**US15 — As a user, I want sentiment analysis results to be stored and retrievable so that they can power mood history visualizations.**

**Acceptance criteria:** *Sentiment label and score persist after analysis; failed analysis does not overwrite good data; mood history endpoint returns stored data including pending entries*

| TC ID | Scenario | Type | Status | Description | Input / Payload | Expected Result | Notes |
|---|---|---|---|---|---|---|---|
| TC_US15_01 | Valid flow | Integration | PASS | After successful analysis the journal entry stores sentiment label and confidence score | Valid submission + successful ML response | DB record contains sentiment_label confidence_score analysed_at | Persistence check |
| TC_US15_02 | System response | Integration | PASS | Stored sentiment data is returned by mood history endpoint | Authenticated GET mood history request | HTTP 200; returned data matches stored DB values | Retrieval correctness |
| TC_US15_03 | Edge case | Integration | PASS | Entry without completed analysis is still returned in history | Entry stored with null sentiment fields | HTTP 200; item returned with Pending label or null-safe handling | Pending state handling |
| TC_US15_04 | System response | Integration | PASS | Failed analysis does not overwrite previously successful sentiment data | Existing analysed entry + later failed update path | Previously saved label/score preserved | Data safety test |
| **Totals** | | | 4 | 0 | 0 | | |

Detailed spreadsheet here

# SPRINT 2 TEST CASE

**Sprint 2**

**US6 — As a user, I want to view my mood history over time so that I can reflect on emotional trends.**

**Acceptance criteria:** *Mood history endpoint returns authenticated user data; chart updates after new submission; empty history handled; unauthenticated access rejected*

| TC ID | Scenario | Type | Status | Description | Input / Payload | Expected Result | Notes |
|---|---|---|---|---|---|---|---|
| TC_US6_01 | Edge case | Integration | PASS | Authenticated user with no analysed entries requests mood history | Valid JWT + no mood history | HTTP 200; empty array returned | Empty history handling |
| TC_US6_02 | Valid flow | Integration | PASS | Authenticated user with one analysed entry requests mood history | Valid JWT + one stored entry | HTTP 200; one item with date sentiment_label confidence_score | Single entry case |
| TC_US6_03 | Valid flow | Integration | PASS | Authenticated user with multiple analysed entries requests mood history | Valid JWT + multiple stored entries | HTTP 200; ordered list of mood history items | Multiple entries case |
| TC_US6_04 | Invalid input | Integration | PASS | Unauthenticated user requests mood history | No token | HTTP 401; unauthorized response | Auth protection |
| TC_US6_05 | System response | Integration | PASS | After new journal submission mood chart data refreshes without full reload | Valid submit followed by history refresh | New data point appears in chart state/API response | Frontend re-fetch/update behavior |
| **Totals** | | | 5 | 0 | 0 | | |

Detailed spreadsheet here

# SPRINT 2 TEST CASE SUMMARY

| Category | Total cases | Passed | Failed |
|---|---|---|---|
| US6 Mood History | 5 | 5 | 0 |
| US11 Registration | 4 | 4 | 0 |
| US12 Login and Auth | 5 | 5 | 0 |
| US13 Past Entries | 5 | 5 | 0 |
| US15 Sentiment Persistence and Stored Mood Data | 4 | 4 | 0 |
| US16 Error Handling | 5 | 5 | 0 |
| **Total** | **28** | **28** | **0** |

# SPRINT 2 STORIES COMPLETED

- **US6 -** Mood History Visualization
- **US11 -** User Registration
- **US12 -** User Login
- **US13 -** View Past Journal Entries
- **US15 -** Store Sentiment Results
- **US16 -** Error Handling
- **US17 -** Loading Indicator
- **US18 -** Improved Mood Visualization
- **US19 -** Authentication SecuritY



Detailed spreadsheet here

# SPRINT 2 STORIES NOT COMPLETED

**These stories will be carried over to Sprint 3 for completion:**

- **US20** – Therapist Access
- **US21** – Enable Sharing
- **US22** – Disable Sharing
- **US23** – Therapist User List
- **US24** – Therapist Dashboard

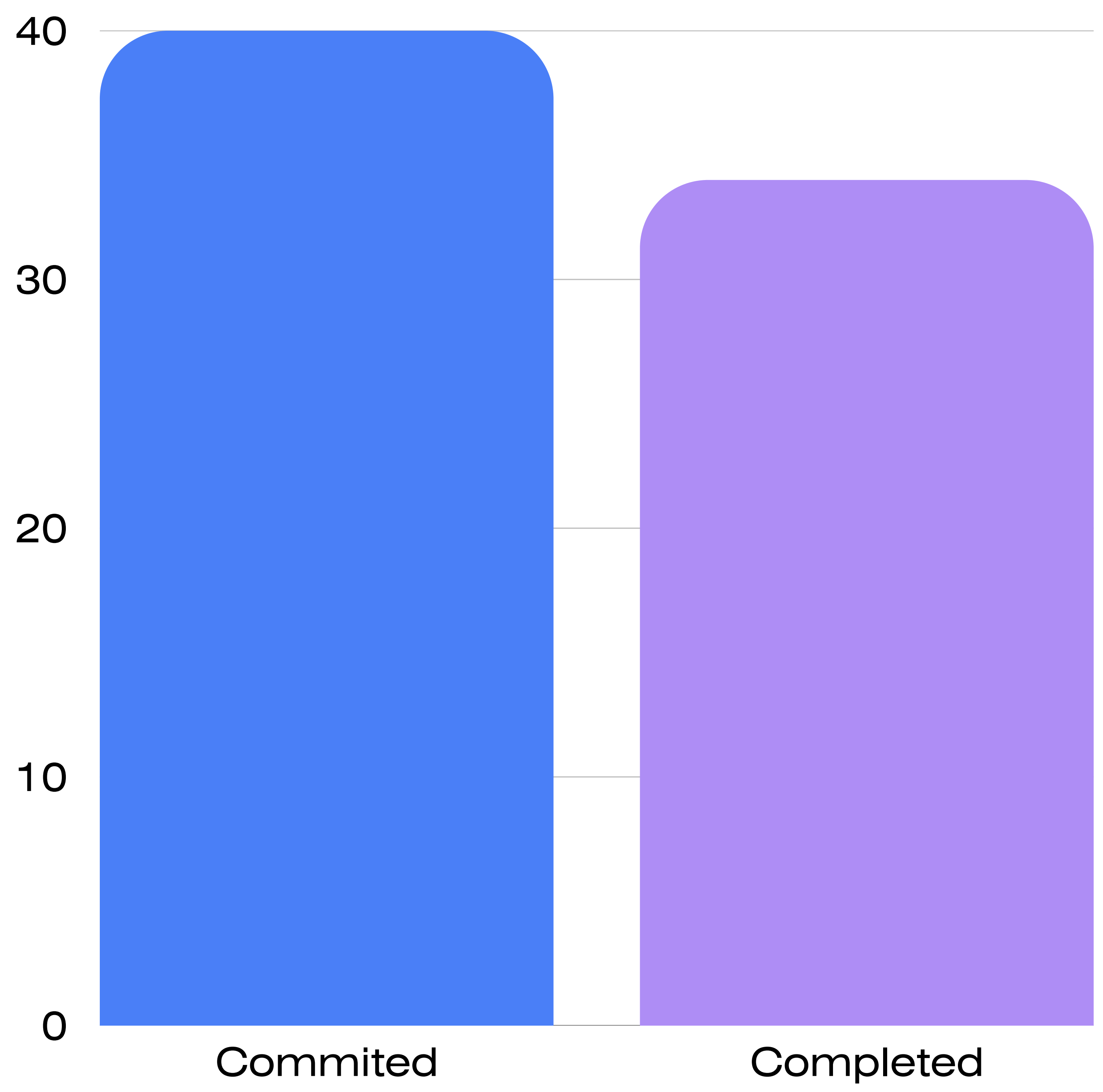


Detailed spreadsheet here

# METRICS

## Team Velocity

| | |
|---|---|
| Team estimated points | **40** |
| Team points delivered | 34 |
| Total points carried over | 6 |
| Team Velocity | 34 |



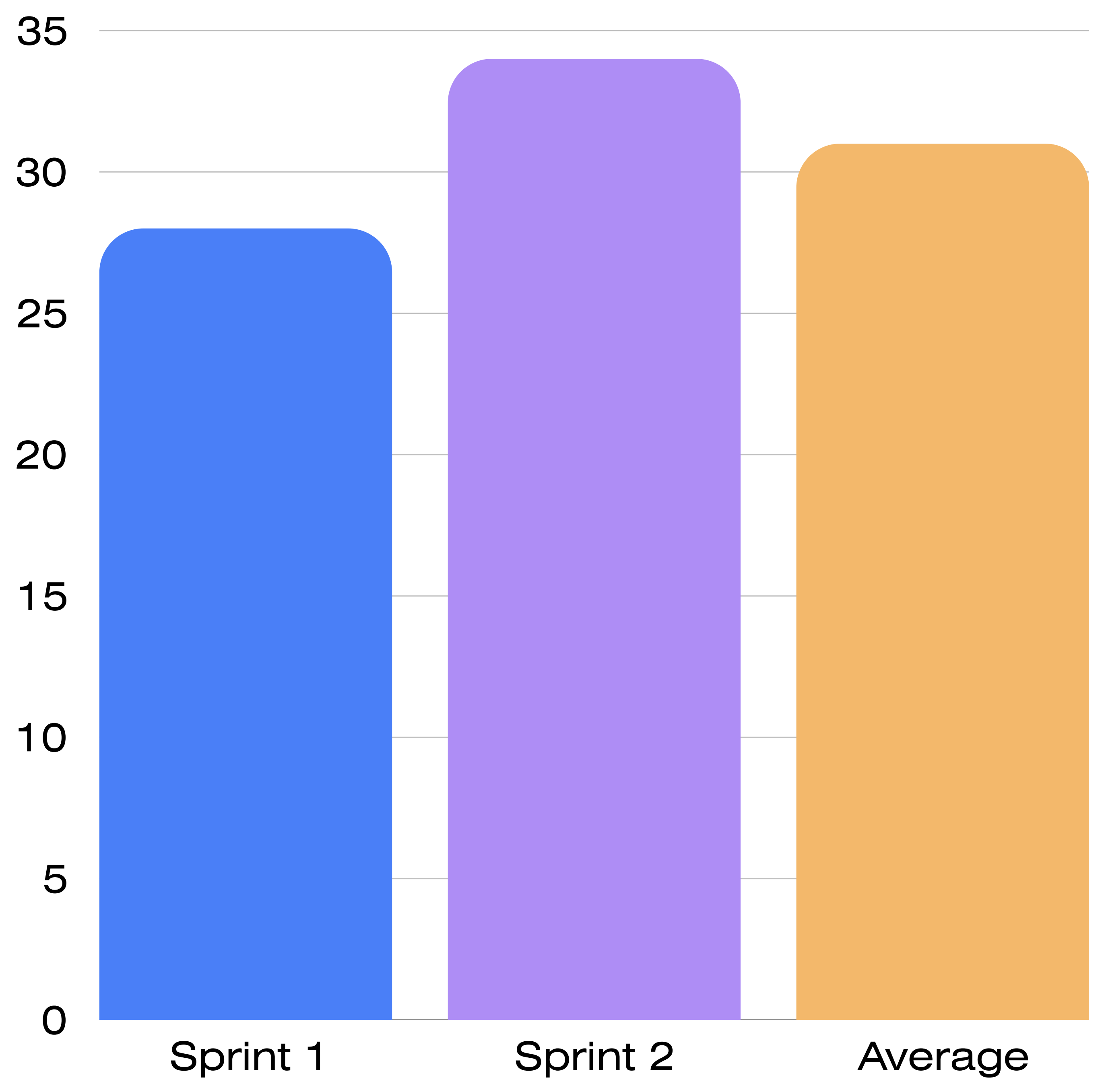
# METRICS

## Team Velocity

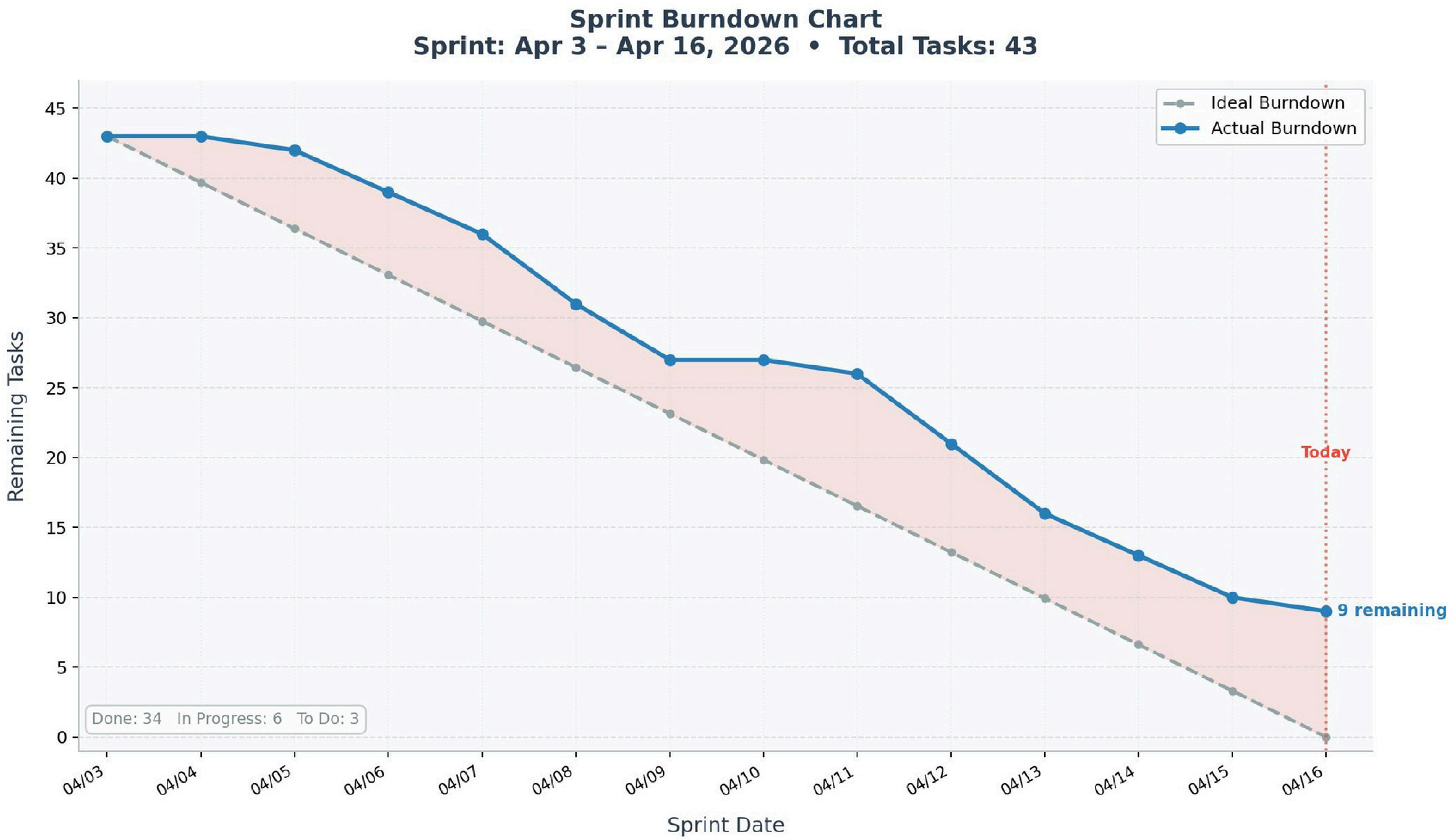
Total velocity of sprint 1: 28

Total velocity of sprint 2: 34

Average: 31



# BURNDOWN STATUS



# COMPLETED VS COMMITTED

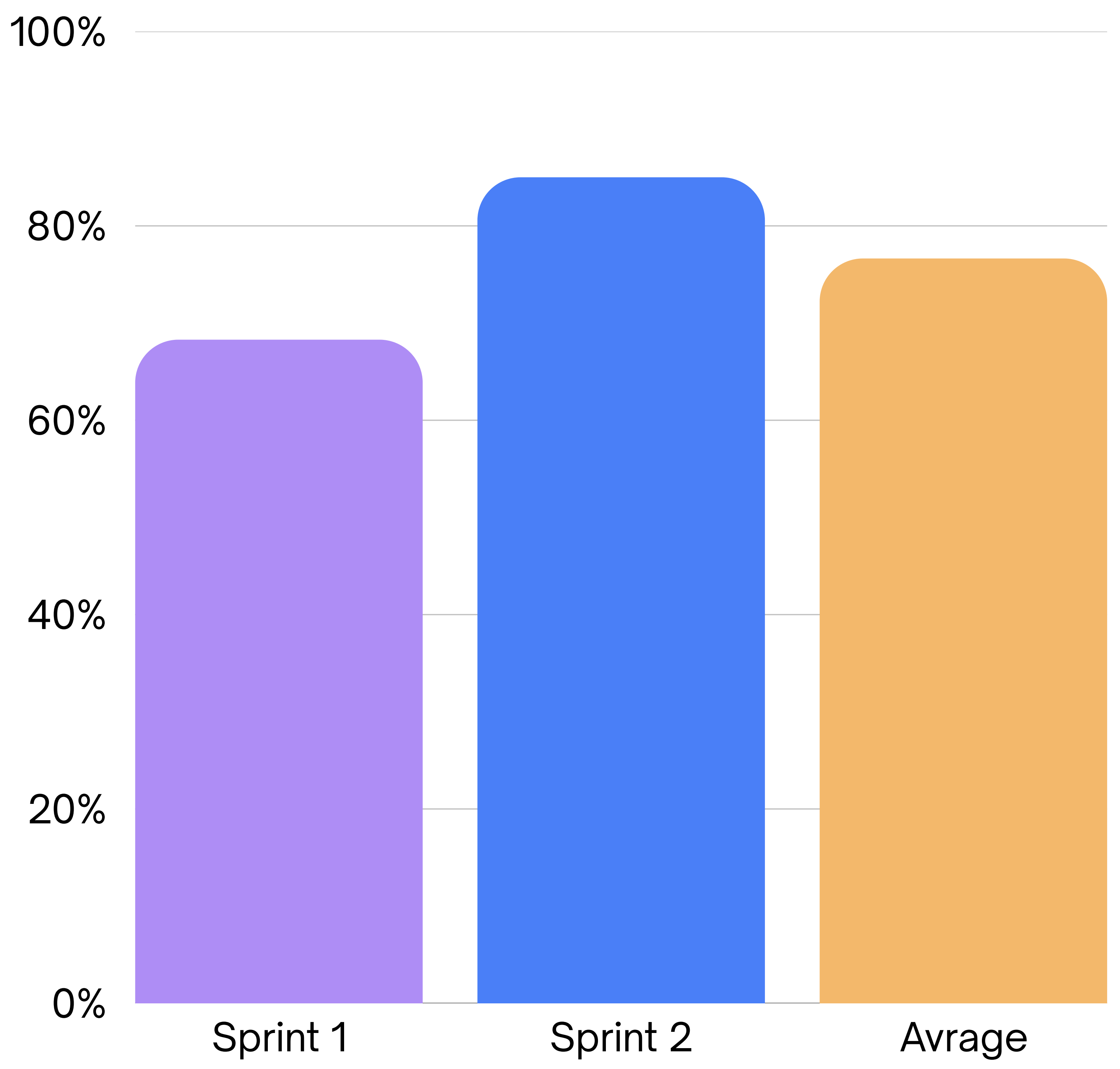
Sprint 1:

- Committed: 41
- Completed: 28
- Ratio: 68.29%

Sprint 2:

- Committed: 40
- Completed: 36
- Ratio: 85%

Average: 76.65%



# RETROSPECTIVE

## WHAT WENT WELL

Daily stand-ups are more structured and focused on blockers and progress

Better collaboration between co-owners on shared tasks

Team members supported each other when blocked

## WHAT COULD BE IMPROVED?

Earlier start on documentation and slides

Communication could be more consi-stent across the team

Improve sprint planning accuracy

## ACTIONS FOR NEXT SPRINT

Begin documentation alongside development

Increase activity on communication platform

Improve estimation during sprint planning

# WIKI PAGE LINK

# DEMO

MIND MIRROR

CHECK IN. REFLECT. GROW.

# THANK YOU

17 April, 2026